Benchmarking Machine Learning and Neural Network Models for Fake News Detection: A Comparative Study

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Introduction

- Fake news undermines public trust, disrupts democracy, and spreads misinformation.
- Bangla fake news detection is crucial due to the language's increasing digital footprint.
- This study explores ML and Neural Network approaches for identifying Bangla fake news.

Objective

- Detect and classify Bangla news as real or fake.
- Use NLP and classification models for text analysis.
- Develop a real-time, user-friendly web interface for prediction.
- Benchmark various ML and neural models for performance comparison.

Literature Review

Name of Paper	Method	Accuracy	Limitation
Multi-modal Robustness Fake News Detection with Cross-Modal and Propagation Network Contrastive Learning.	Multi-modal Robustness Fake News Detection with Cross-Modal and Propagation Network Contrastive Learning	87.71%	The model focuses on text, images, and propagation networks but does not incorporate video or audio, which are common in misinformation.
Cross-attention multi-perspective fusion network based fake news censorship.	Cross-Attention Multi-Perspective Fusion Network Based Fake News Censorship.	89.5%	This model only for English and chines language.
GBERT: A hybrid deep learning model based on GPT-BERT for fake news detection.	GBERT: A hybrid deep learning model based on GPT-BERT for fake news detection.	95.31%	BERT and GPT require significant computational resources, making real-time implementation challenging.
Cross-lingual COVID-19 Fake News Detection.	CrossFake framework using BERT, crosslingual detection.	75%	Translation quality affecting model performance, difficulty in capturing information in long texts, small dataset size.
Fake news detection on social networks using Machine learning techniques.	Naïve Bayes ,Support Vector Machine (SVM), Random Forest.	93%	Research is needed to improve detection on multilingual datasets and real-time fake news identification.

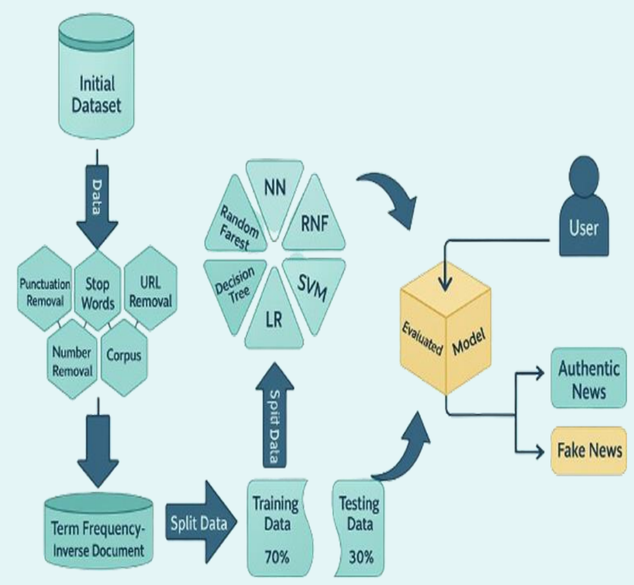
Methodology

- **Data Collection**: 3,045 Bangla news articles from Prothom Alo, BDNews24, Jugantor, and Bangla Tribune.
- **Preprocessing**: Cleaning, tokenization, lemmatization, TF-IDF vectorization.
- **ML Models Used**: AdaBoost, Decision Tree, Gradient Boosting, KNN, Logistic Regression, Naive Bayes, Random Forest, SVM and XGBoost.
- **Evaluation Metrics**: Accuracy, Precision, Recall, F1-score, CPU/Memory usage.

Outcome

- Gradient Boosting was the top performer with **83.15% accuracy** and **87.64% F1-score**.
- Logistic Regression, Support Vector Machine (SVM) and Random Forest also performed consistently well.
- A simple React-based frontend enables real-time prediction for users.
- TF-IDF and word embeddings significantly boosted classification performance.

Methodology Block Diagram



Conclusion

- ML-based fake news detection is viable and effective for Bangla news.
- Ensemble models like Gradient Boosting are highly accurate and reliable
.
- Word embeddings and vectorization enhance semantic understanding.
- The project delivers both a model and a functional web interface.

Limitations

- Language ambiguity in Bangla can mislead models.
- Data imbalance between real and fake news affects learning.
- Lack of multilingual NLP tools for Bangla text.
- Domain-specific news may not generalize well.
- Real-time prediction with high accuracy is challenging.